Fine-Tuned Open-Weight Language Models for Explainable Energy-Theft Detection: Accuracy, Generalization, and Deployment Cost

Emran Altamimi, Abdulaziz Al-Ali, Abdulla K. Al-Ali, and Qutaibah M. Malluhi

***Abstract*—Electricity-theft detection from smart-meter data has been dominated by classifiers that return a flag and nothing else, and the frontier language models that can now also explain a flag are proprietary: every query is billed, consumption data leaves the utility, and the model cannot be retrained as manipulation patterns change. This paper asks whether locally fine-tuned open-weight models close that gap. We fine-tune seven instruction-tuned models spanning 4B to 31B parameters with one shared low-rank adaptation recipe on a thirteen-class theft-identification task and evaluate them under a consumer-disjoint protocol against the zero-shot proprietary frontier and matched classical baselines. Fine-tuning lifts the best open model to a macro-F1 of 0.9331 against the proprietary frontier's 0.929, and an 8B model is statistically tied with it. A random forest reaches 0.9122 on the identical split, so accuracy alone justifies the language model only weakly; its durable advantages are specificity, 0.892 against the forest's 0.715, which halves false alarms and therefore site inspections, and the content it attaches to each flag. The fine-tuned model names the manipulation mechanism, scored 4.29 of 5 by a cross-family LLM judge, estimates the stolen fraction to a mean absolute error of 0.0412 against the exact pre-attack quantity, and localizes the manipulated window. Detection transfers to ten of twelve attack types never seen in training; identification does not. The results support a deployment in which the classical model screens every meter-day and the fine-tuned model adjudicates the flagged residue, with the fine-tuning cost repaid within roughly 400 query-days relative to a proprietary API.**

***Index Terms*—Energy theft, non-technical losses, large language models, fine-tuning, smart grid, explainability.**

I. Introduction

Non-technical losses (NTL) from electricity theft cost grid utilities tens of billions of dollars annually and remain one of the largest sources of unbilled energy in distribution systems [1]. Advanced metering infrastructure has transformed electricity distribution by enabling fine-grained, user-level consumption monitoring at scale [2]. Together, these have made automated detection directly from smart-meter consumption data an active and sustained research direction.

NTL detection has been dominated by two families of methods: supervised classifiers trained on hand-engineered load-profile features, spanning gradient-boosted trees, ensemble methods, and deep convolutional networks, and unsupervised novelty detection for circumstances where labeled theft examples are unavailable [3]–[6]. Both families either return a bare label or an anomaly score with no account of why a day or consumer was flagged, which limits their operational value to a utility provider: an inspector receiving a bare classification cannot distinguish a meter bypass, which requires a physical field visit, from a false-data-injection attack, which requires a cybersecurity response. Prior work showed that a proprietary frontier LLM can outperform these classical baselines on fine-grained, 13-class theft identification without any task-specific training [7], but a proprietary API is not a deployable detector: every query is billed at expensive API rates, consumption data leaves the utility's infrastructure, which raises a privacy concern, and the model cannot be retrained as new manipulation patterns emerge. Whether an open-weight model, fine-tuned locally, can match that zero-shot ceiling remains unanswered, and so do the questions that follow from it: at what data and compute cost, with what robustness to unseen attacks, and with what capacity to explain its own verdict.

This paper answers that question directly. We fine-tune seven open-weight instruction-tuned models, spanning 4B to 31B parameters across five model families, with a single shared low-rank adaptation recipe, and evaluate them against the zero-shot proprietary frontier and a set of classical baselines under an identical, consumer-disjoint evaluation protocol. Beyond classification, we ask whether a fine-tuned model can also justify its verdict in human-understandable text, naming the manipulation mechanism and estimating its magnitude to determine severity, and we adopt a rationale-distillation pipeline whose teacher diagnoses each day from the readings alone, revealing the label only where that diagnosis fails, and we verify the resulting explanations against the attacks' recorded generative parameters, so that explanation quality is measured against ground truth rather than against textual fluency. The explanations are further scored by an independent LLM judge, deliberately drawn from a different model family than the teacher, to avoid the self-preference bias that same-family judging is known to introduce.

This paper makes the following contributions:

- A controlled screening of seven open-weight instruction-tuned LLMs, spanning 4B to 31B parameters across five model families, fine-tuned under a single shared low-rank adaptation recipe and benchmarked against a zero-shot proprietary frontier and a set of classical baselines on 13-class energy-theft identification.
- A data- and compute-efficiency analysis that identifies the

The authors are with the Department of Computer Science and Engineering and the KINDI Center for Computing Research, Qatar University, Doha, Qatar. Corresponding author: Emran Altamimi (e-mail: ea1510662@qu.edu.qa).

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point at which the marginal value of an additional labeled training day collapses, establishing a practical labeling budget for utilities adopting this approach.
- A generalization study that withholds entire attack families from training and measures detector survival on manipulation types never seen during fine-tuning.
- A severity-stratified failure analysis that characterizes detection performance as a function of the fraction of energy actually removed, rather than treating all theft instances as equally detectable.
- A rationale-distillation evaluation that checks each explanation's stated numeric parameter against the attack's recorded generative parameters, so that explanation correctness is measured against ground truth rather than by textual overlap alone.
- A bias-controlled LLM-as-judge protocol, drawing the judge from a different model family than the rationale-generating teacher to avoid self-preference bias, that evaluates the resulting free-text explanations for behavioral correctness, completeness, faithfulness to the data, and coherence.
- A cost–performance deployment analysis comparing fine-tuned open-weight models, the zero-shot proprietary frontier, and classical baselines under a common operating point, identifying where a locally fine-tuned detector becomes the preferable deployment choice.

II. Related Work

This section reviews prior work on energy-theft detection and its adaptation to large language models, then turns to the fine-tuning and evaluation methodology this paper builds on.

Machine-learning methods for non-technical-loss detection have been extensively studied and divide broadly into supervised and unsupervised approaches [2], [8]. The majority of this literature consists of supervised classifiers: Punmiya and Choe applied feature-engineered gradient boosting, Yan and Wen used extreme gradient boosting in advanced-metering-infrastructure settings, and Gunturi and Sarkar validated random-forest ensembles [3]–[5]. Deep architectures have also progressed from an early wide-and-deep convolutional network to more recent hybrid designs combining convolutional, attention, and autoencoder components [6], [9], [10]. To address the scarcity of verified theft labels, a parallel line of work turns to unsupervised or semi-supervised detection: Viegas et al. applied clustering-based novelty detection, Messinis et al. proposed a hybrid that combines anomaly scoring with supervised refinement, and a separate strand compares observed consumption against a forecasted or reference baseline using residual or distributional dissimilarity measures [11]–[13]. These unsupervised methods reduce labeling requirements but generally achieve lower detection accuracy than their supervised counterparts [2]. Recent work has begun pairing such detectors with post-hoc tools such as SHAP for feature-level justification [14]. Across both families, the output remains a label, score, or feature-attribution map with no description of the underlying manipulation mechanism, the gap this paper's explanation objective is designed to close.

A recent separate line of work applies large language models to numerical and time-series reasoning more broadly. Zhou et al. showed that a frozen pretrained language model can serve as a general-purpose backbone across forecasting, classification, and anomaly detection, and Gruver et al. demonstrated that numerical time series can be serialized directly as text for LLM inference without architecture changes [15], [16]. Alnegheimish et al. extended this to zero-shot LLM anomaly detection, and Hegselmann et al. showed that serializing tabular rows to text and fine-tuning an LLM classifier outperforms gradient-boosted trees in low-data regimes [17], [18]. Within the smart-grid domain, Selim et al. applied an LLM directly to textual power-system control commands for cyberattack detection, and Saber et al. fine-tuned compact LLMs on textualized relay current measurements to distinguish cyberattacks from genuine faults [19], [20]. Closest to the present study, our own earlier work showed that a zero-shot proprietary frontier LLM can outperform supervised classical baselines on fine-grained, 13-class energy-theft identification with no task-specific training [7], leaving open whether a fine-tuned, locally deployable open-weight model can reach the same ceiling while also producing a free-text explanation of its verdict, rather than the attention-based or feature-level interpretation the earlier work relies on.

Adapting a large model to a specialized task without full retraining is now standard practice through parameter-efficient fine-tuning: LoRA constrains weight updates to a low-rank subspace, DoRA decomposes weights into magnitude and direction for improved adaptation, and QLoRA combines low-rank adaptation with 4-bit quantization to make fine-tuning feasible on a single GPU [21]–[23]. Beyond adapting parameters, a separate body of work distills a large teacher's reasoning into a smaller student. STaR bootstraps rationale generation by prompting a model to explain problems it solves correctly; when its initial attempt is wrong, it falls back to a rationalization step that reveals the answer and asks the model to reason backward from it [24]. Fine-tune-CoT generates reasoning samples from a large teacher and keeps only those whose final prediction is correct, using the surviving rationale-answer pairs as fine-tuning data [25]. A more recent line of work verifies the rationale itself before using it as supervision, rather than filtering by final-answer correctness alone: program-aided distillation checks whether a teacher-generated program executes to the correct answer, and a comparable approach verifies programmatic reasoning in vision-language distillation [26], [27]. These methods verify a teacher-generated program by executing it and checking its output; this paper extracts the corresponding numeric claim directly from a free-text rationale and checks it against the attack's recorded generative parameters, applying the check as an evaluation instrument rather than as a filter on the training supervision.

Evaluating free-text explanations at scale has motivated using a large language model itself as an automated judge. Zheng et al. established this protocol and showed that GPT-4 judgments agree with human preference at a rate comparable to inter-human agreement, while also identifying position, verbosity, and self-enhancement biases that can distort a judge's scores [28]. Liu et al. showed that reference-based metrics such as ROUGE and BLEU correlate poorly with human judgment on generation quality, and that an LLM judge prompted to reason before scoring achieves substantially higher correlation with human raters, motivating a judge-based protocol alongside, rather than in place of, lexical overlap metrics [29]. Self-enhancement bias in particular has been shown to be systematic: judges recognize and favor their own outputs, and this preference intensifies after fine-tuning, which is the basis for drawing this paper's judge from a different model family than its rationale-generating teacher [30]. Judge quality is one axis of a deployable detector; its cost is the other. Cascade architectures that route easy cases to a cheap model and escalate only uncertain ones to a more expensive model have been shown to match a frontier model's accuracy at a fraction of its inference cost [31], a pattern this paper's cost–performance analysis extends to a classical screen paired with LLM adjudication.

III. Experimental Setup

Prior work established that a zero-shot proprietary model can outperform supervised classical baselines on fine-grained energy-theft identification [7]. However, a proprietary API is an impractical operating point for a utility provider. Every query-day is billed at API rates, consumption data leaves the utility's infrastructure, and the detector cannot be adapted when new manipulation patterns emerge. This section asks whether small open-weight models, fine-tuned locally, can close that gap, and at what cost. We answer six research questions:

1) **Frontier match (RQ1):** Can fine-tuned open-weight models at locally deployable scale, 4B to 14B parameters with a 31B model as the upper reference, match the zero-shot proprietary frontier and supervised classical baselines on 13-class theft identification?
2) **Label and compute cost (RQ2):** If open models match the frontier, how many labeled days and how much training compute does it require, and where does performance saturate?
3) **Generalization (RQ3):** If the label bill is affordable, does the detector survive attack types that were absent from its training data?
4) **Severity (RQ4):** Where does any detector fail, as a function of the fraction of energy actually removed?
5) **Explanation (RQ5):** Beyond detection, can the model state the manipulation mechanism and quantify its magnitude, and at what cost to classification?
6) **Deployment (RQ6):** Given the answers above, what does the cost–performance frontier look like against API zero-shot and classical ML, where is the break-even, and what operating workflow do the results recommend?

The remainder of this section specifies the shared apparatus: the dataset and its frozen partition, the attack taxonomy, the models and their single shared recipe, the baselines, the metrics, and the statistical protocol. The design logic that connects the experiments is stated last, in Section III-F.

A. Task, Dataset, and Data Partitioning

The task is per-day classification of smart-meter consumption. Given one query day of readings together with the consumer's preceding reference week, the model emits a structured output that declares the day *original* or *manipulated* and, when manipulated, identifies which of twelve attack mechanisms produced it. The model input is text-serialized numeric data. Each sample contains a metadata line (consumer, month, week, and day identifiers), per-day summary statistics (minimum, maximum, mean, and standard deviation), and calendar features (month, day-of-week, day-of-month). It then carries the raw readings: 96 fifteen-minute values for each of the seven reference days and the query day, 768 numbers in total. No features beyond these are added.

All experiments draw from a single dataset of 8,239 labeled consumer-days spanning 324 consumers and all 13 classes. We partition it once, before any experiment, into a consumer-disjoint 80/20 split. The 259 training consumers contribute 6,589 days; the remaining 65 consumers contribute the 1,650 evaluation days. Consumer disjointness matters because per-consumer consumption habits are strong identifiers. A random row-level split would leak consumer identity from train to test and inflate the reported metrics. The evaluation slice contains 130 benign days and 1,520 manipulated days (105–130 per attack class), an 11.7:1 imbalance. Every experiment in this paper uses this one partition.

During fine-tuning, each training example pairs the serialized readings with the correct structured answer, which comprises the binary verdict, the attack identification, and the day's reference explanation (Section III-E). The loss is computed on the answer tokens only; the model is never trained to reproduce its ∼7,900-token numeric input. At evaluation, the model decodes deterministically and its answer is parsed for scoring.

B. Attack Taxonomy

Table I lists the 13-class taxonomy: the benign class plus twelve synthetic attack mechanisms drawn from the non-technical-loss literature [2], [32]. Each mechanism is a distinct transformation of the daily load profile. After injection, every manipulated reading is re-quantized to the meter's native reading step, so a floating-point transformation cannot leave off-grid values that would offer the detector a shortcut no physical tamper produces. The taxonomy spans the practically relevant spectrum, from total meter bypass (attack0) to manipulations that leave total energy unchanged and alter only the profile's shape (attack10). For the generalization experiments of Section IV-C we group the twelve mechanisms into four families by manipulation type: zero-suppression, multiplicative or randomized reduction, threshold clipping or subtraction, and shape substitution.

C. Models and Fine-Tuning Configuration

We screen seven open-weight instruction-tuned models spanning 4B to 31B parameters and five model families: Qwen3-8B and Qwen3-14B [33], Phi-4-reasoning (14B) [34],

TABLE I

THE 13-CLASS TAXONOMY: THE BENIGN PROFILE PLUS TWELVE ATTACK MECHANISMS, WITH THE FAMILY GROUPING USED FOR THE OUT-OF-CLASS ROTATIONS OF SECTION IV-C.

Class	Mechanism	Effect on profile	Family
original	unmodified consumption	—	—
attack0	all samples set to zero for the whole day	removes trend and magnitude	Z
attack1	all samples $\times$ one random factor $\in (0,1)$	trend kept, magnitude reduced	M
attack2	a random contiguous window zeroed (on–off)	changes trend and amplitude	Z
attack3	each sample $\times$ its own random factor $\in (0,1)$	changes trend and amplitude	M
attack4	readings reduced inside a random sub-period	changes trend and amplitude	M
attack5	each sample $\leftarrow$ mean consumption $\times$ random $\in (0,1)$	replaces the pattern	M
attack6	values above a random cut-off clipped to it	changes trend and amplitude	T
attack7	a random cut-off subtracted; negatives to zero	trend kept, total reduced	T
attack9	every sample $\leftarrow$ the day's average (flat profile)	removes all trend	S
attack10	the daily pattern reversed in time	trend changed, total unchanged	S
attack12	profile of a lower-consuming user substituted	false data injection	S
attack13	each interval reports the true value or zero	irregular; trend and amplitude	Z

Families: Z = zero-suppression, M = multiplicative/randomized, T = threshold, S = shape substitution.
attack13 is the IEEE sampled-zeroing attack; attacks 8 and 11 of the source taxonomy are not present in the dataset.

TABLE II

THE FROZEN FINE-TUNING RECIPE SHARED BY EVERY RUN IN THIS PAPER. ONLY THE AXIS UNDER STUDY VARIES PER EXPERIMENT: THE BASE MODEL (RQ1), THE TRAINING-SET SIZE (RQ2), THE HELD-OUT ATTACK FAMILY (RQ3), THE OUTPUT TARGET (RQ5), AND THE LORA RANK (APPENDIX A).

Knob	Value
Adapter	LoRA (DoRA), $r = 16$, $\alpha = 32$, dropout 0.05
Target modules	all linear layers
Quantization	bf16 (31B model: 4-bit NF4 QLoRA)
Epochs / batch	1 epoch, batch 1 $\times$ 8 gradient-accumulation steps
Learning rate	2×10^{-4}, cosine schedule, 3% warmup
Weight decay	0.01
Max sequence length	8,192–11,008 tokens (tokenizer-dependent)
Decoding at evaluation	greedy, 512 new tokens
Hardware	one NVIDIA H100 80 GB

Ministral-3-14B [35], Llama-3.1-8B [36], Gemma-4-E4B (4B effective), and Gemma-4-31B [37]. Every model is fine-tuned with the identical recipe of Table II, LoRA [21] in its weight-decomposed variant [22]; only the base model varies. This single-recipe design is a deliberate trade-off. It makes the screening comparison clean at the risk of disadvantaging models whose optimal recipe differs, and we validate the choice retrospectively in Appendix A. The 31B model exceeds the memory budget for 16-bit adaptation on a single GPU and is trained with 4-bit quantized adaptation [23] instead; we flag this asymmetry where it matters.

Two evaluation choices require explanation. First, after training, the low-rank adapter weights are folded back into the base model, so each evaluated model is a single standalone checkpoint, the same artifact a utility would deploy, and the reported inference throughput refers to that artifact. Second, a generative classifier outputs a discrete label rather than a continuous score, which would leave threshold-based metrics such as AUC undefined. We recover a score from the model itself: with the output held fixed up to the classification field, the model's probabilities for the two class words are renormalized into a probability of manipulation. This restores AUC and threshold analysis without altering any generated prediction.

D. Baselines

Two kinds of comparison appear throughout. The first is the *zero-shot proprietary frontier*, Gemini 3.1 Pro, evaluated on the same evaluation slice (0.976 binary balanced accuracy, 0.929 macro-F1). The second is a set of *classical baselines* trained on the same training split and evaluated on the same slice: a random forest, CatBoost, XGBoost, a logistic regression, and an unsupervised local outlier factor, all consuming a compact feature vector computed from the same readings and reference week the LLMs receive (energy and shape ratios of the query day against the reference week, dispersion, near-zero fraction, and quarter-day means). These support paired statistical comparisons. CatBoost and XGBoost carry the full metric set; the random forest is recorded with balanced accuracy, specificity, and identification macro-F1. The logistic regression and the local outlier factor provide binary-detection context only, the former because it was fit as a plain binary detector and the latter because an unsupervised outlier score has no thirteen-class output; both are therefore reported as balanced accuracy alone.

E. Evaluation Metrics and Statistical Protocol

Classification metrics: Binary detection is reported as balanced accuracy, F1 on the manipulated class, AUC-ROC (via verbalizer scores), specificity, precision, and recall. Identification is reported as macro-F1, weighted-F1, balanced accuracy, and per-class F1 over the 13 classes. We emphasize specificity alongside recall because the two error types carry asymmetric operational cost. A false positive is a site inspection dispatched to an honest consumer, and at utility scale the binding constraint is usually the inspection budget rather than the detection rate.

Statistical protocol: All headline metrics carry 95% percentile confidence intervals from a cluster bootstrap [38] that resamples the 65 evaluation consumers with replacement ($B = 10{,}000$, one fixed seed throughout). Days from the same consumer share consumption habits and are therefore not independent; resampling whole consumers preserves this dependence, where a per-day bootstrap would understate the interval width. Paired model comparisons combine the same bootstrap, applied to the metric difference, with a McNemar test on the per-day predictions [39]. The classical test treats discordant days as independent, which they are not here: every attacked day is derived from a clean source day of one consumer, so discordant outcomes cluster. All McNemar p-values in this paper therefore use the cluster-adjusted variant of Durkalski et al. [40], which estimates the variance of the discordant-pair difference at the consumer level and reduces to the classical statistic when each consumer contributes a single day. When a family of comparisons is tested together, the family-wise error rate is controlled with Holm's correction [41].

Explanation metrics: Each day in the dataset carries a reference rationale written by a teacher model, Gemini 3 Pro, under the standard self-taught-reasoning protocol [24]. The teacher diagnoses the day from the readings alone, without the ground-truth label; only where its diagnosis is wrong is the label revealed and the rationale regenerated (the *rationalization* fallback, about 11% of days). Rationale quality is measured by ROUGE-L [42] against this reference, more stringently by parameter-extraction fidelity, and at the level of whole explanations by an independent LLM-as-judge protocol [28] whose design and results accompany the rationale experiment (Section IV-E). Five of the twelve mechanisms are governed by a single recorded value: the retained fraction for attack1, the zeroed window for attack2, the scaled sub-period for attack4, and the clipping or subtraction cut-off for attack6 and attack7. For these we check whether the explanation states the value that actually generated the attack. The remaining seven mechanisms draw an independent random value per reading or replace the profile wholesale, so no single parameter exists to restate. Magnitude estimates (Section IV-F) are scored by mean absolute error on the energy-reduction fraction against the pre-attack day, alongside rank correlation, bucket accuracy, and window intersection-over-union.

Severity proxy: Several analyses stratify by theft severity using $\rho = 1 - E_{\text{query}}/\overline{E}_{\text{reference week}}$, the fraction of expected energy removed, with bucket edges at 10%, 30%, and 60%. We measure severity against the reference week rather than against the pre-attack profile of the same day for two reasons. The counterfactual clean day is unobservable in deployment, whereas the reference week is exactly the context the detector receives, so this proxy is computable identically in evaluation and in production. It is also defined for benign days, which the same-day comparison is not, and this yields the calibration that validates the bucket edges: benign day-to-day variation of ρ spans -0.197 to $+0.178$ (5th to 95th percentile), so the 30% edge sits outside natural variation. Because the attacks are synthetic, the pre-attack day itself is nonetheless on record, and the magnitude estimates of Section IV-F are scored against that exact quantity; the proxy is the definition the severity buckets use. Two mechanisms, attack9 and attack10, do not change the day's total energy; they only reshape the consumption profile (median $\rho \approx 0$). We call these the *energy-neutral* pair and report them separately rather than forcing them into severity buckets.

F. Experimental Design and Research Questions

A single screening pass (RQ1) fine-tunes all seven models once under the shared recipe. From its results, two reference models are selected by a rule fixed before the screening ran: the *flagship* is the model with the best macro-F1, and the *workhorse* is the smallest model whose paired-bootstrap 95% confidence interval on the macro-F1 gap to the flagship contains zero, with lower inference cost as the tie-break. All further fine-tuning, covering the data-efficiency curve (RQ2), the unseen-attack rotations (RQ3), the rationale and magnitude extensions (RQ5), and the rank validation of Appendix A, is carried out on the workhorse alone. The flagship re-enters in the severity analysis (RQ4) and the deployment frontier (RQ6), both of which re-use the screening predictions. **[EDIT: Experiment-dependency diagram (screening $\rightarrow$ selection $\rightarrow$ workhorse experiments $\rightarrow$ deployment): TikZ or draw.io; the draft ships without it.]**

IV. Results and Discussion

In this section, we present and discuss the results. The screening pass answers whether fine-tuned open models reach the zero-shot frontier and selects the workhorse model that all further training runs on. We then cost that result in labels and compute, and stress it against attack families the model never saw. The failure mechanism exposed by that stress test is formalized into a severity analysis, which sharpens the question of what an LLM offers that a much cheaper classical model does not; the rationale-distillation and magnitude experiments answer it. The deployment frontier then assembles accuracy, cost, and capability into an operating recommendation.

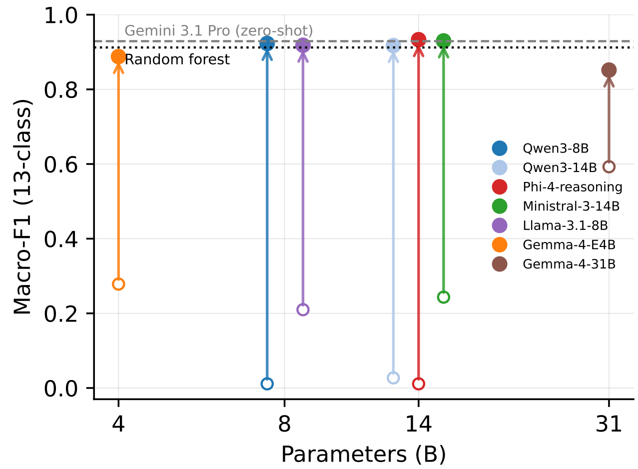

Fig. 1. Macro-F1 against parameter count before and after fine-tuning. Each arrow joins a model's zero-shot score to its fine-tuned score; same-size models are dodged horizontally for legibility. The dashed band is the zero-shot Gemini 3.1 Pro score and the dotted band the random forest baseline.

A. Screening and Model Selection (RQ1)

This experiment establishes the premise of the paper: whether fine-tuned open-weight models at practical scales reach the zero-shot proprietary frontier that motivated this work. Table III reports all seven models in both their zero-shot and fine-tuned conditions on the 1,650-day evaluation slice, against the zero-shot frontier and the classical baselines. Figure 1 visualizes the fine-tuning lift per model.

The answer to RQ1 is affirmative. The best fine-tuned model, Phi-4-reasoning, reaches 0.9331 macro-F1, which exceeds Gemini 3.1 Pro's 0.929 on the same split, and four of the seven models land within 0.016 of it. However, the zero-shot rows show that none of this competence is present before fine-tuning, and the failures are instructive because they are not uniform. Qwen3-8B and Phi-4-reasoning predict *original* for essentially every day (F1 of 0.0000 at a specificity of 1.0). Llama-3.1-8B and Gemma-4-E4B collapse in the opposite direction and declare nearly every day manipulated (specificity of 0.0231 and 0.0154). Notably, zero-shot Qwen3-8B attains an AUC of 0.9066 while its F1 is exactly zero: the base model already ranks manipulated days above benign ones and lacks only a usable decision boundary. This suggests that fine-tuning chiefly teaches the model where to place that boundary, an interpretation the data-efficiency curve of Section IV-B will support directly. The single genuine zero-shot competitor is Gemma-4-31B at 0.5927 macro-F1. Yet the same model, fine-tuned, is the *worst* of the seven at 0.8517, receiving the

TABLE III
MAIN RESULTS ON THE 1,650-DAY CONSUMER-DISJOINT EVALUATION SLICE, IN FIVE BLOCKS. ALL ROWS ARE EVALUATED ON THE SAME SLICE; THE CLASSICAL BASELINES ARE TRAINED ON THE SAME TRAINING SPLIT. MODEL ROWS ARE ORDERED BY PARAMETER COUNT. BEST VALUE PER COLUMN IS BOLDED WITHIN THE FINE-TUNED BLOCK ONLY.

	Model	Binary detection				13-class identification		
		Bal-acc	F1	AUC	Specificity	Macro-F1	Weighted-F1	Bal-acc
Zero-shot proprietary	Gemini 3.1 Pro	0.976	—	—	—	0.929	—	—
Zero-shot open	Gemma-4-E4B	0.5041	0.9560	0.8747	0.0154	0.2784	—	0.3842
	Qwen3-8B	0.5000	0.0000	0.9066	1.0000	0.0112	—	0.0769
	Llama-3.1-8B	0.5059	0.9543	0.8425	0.0231	0.2098	—	0.2885
	Phi-4-reasoning	0.5000	0.0000	0.5863	1.0000	0.0112	—	0.0769
	Ministral-3-14B	0.6893	0.5648	0.9473	0.9846	0.2434	—	0.2986
	Qwen3-14B	0.5053	0.0208	0.9193	1.0000	0.0271	—	0.0858
	Gemma-4-31B	0.9137	0.9767	0.9841	0.8615	0.5927	—	0.6515
Fine-tuned	Gemma-4-E4B	0.9456	0.9878	0.9941	0.9077	0.8878	0.8893	0.8850
	Qwen3-8B	0.9406	0.9898	0.9939	0.8923	0.9229	0.9247	0.9210
	Llama-3.1-8B	0.9191	0.9895	0.9939	0.8462	0.9181	0.9197	0.9177
	Phi-4-reasoning	0.9502	0.9924	0.9964	0.9077	**0.9331**	**0.9341**	**0.9321**
	Ministral-3-14B	**0.9569**	0.9921	**0.9972**	0.9231	0.9298	0.9307	0.9287
	Qwen3-14B	0.9541	**0.9928**	0.9947	0.9154	0.9173	0.9188	0.9154
	Gemma-4-31B	0.9556	0.9800	0.9890	**0.9462**	0.8517	0.8537	0.8503
Supervised ML	Random forest	0.8475	—	—	0.7154	0.9122	—	—
	CatBoost	0.8145	0.9765	0.9828	0.6462	0.9093	—	0.9098
	XGBoost	0.8202	0.9751	0.9811	0.6615	0.9076	—	0.9075
	Logistic regression	0.8669	—	—	—	—	—	—
Unsupervised	LOF	0.8209	—	—	—	—	—	—

Dashes mark metrics not recorded for that model.

TABLE IV
SELECTING THE WORKHORSE: PAIRED CONSUMER-CLUSTER BOOTSTRAP (65 CLUSTERS, $B = 10{,}000$) ON EACH MODEL'S MACRO-F1 GAP TO THE FLAGSHIP. THE WORKHORSE ROW IS BOLDED.

Model	Params (B)	Macro-F1	Δ vs. flagship	95% CI	Tied
Phi-4-reasoning	14	0.9331	—	—	flagship
Ministral-3-14B	14	0.9298	-0.0033	$[-0.0141, +0.0072]$	yes
Qwen3-8B	**8**	**0.9229**	$\mathbf{-0.0102}$	$[\mathbf{-0.0225}, \mathbf{+0.0021}]$	**yes**
Llama-3.1-8B	8	0.9181	-0.0149	$[-0.0284, -0.0022]$	no
Qwen3-14B	14	0.9173	-0.0158	$[-0.0286, -0.0036]$	no
Gemma-4-E4B	4	0.8878	-0.0453	$[-0.0576, -0.0337]$	no
Gemma-4-31B	31	0.8517	-0.0814	$[-0.1000, -0.0631]$	no

smallest lift from adaptation. This model is also the only one adapted in 4-bit, so its poor showing conflates scale with quantization and we do not read a scale conclusion from it. The converse case carries no such confound: Phi-4-reasoning enters with the weakest zero-shot signal of all (a verbalizer AUC of 0.5863, barely above chance) and leaves as the flagship. In other words, zero-shot competence and fine-tuned competence are different properties, and neither predicts the other.

Table IV applies the selection rule of Section III-F, fixed before the screening ran: the best model becomes the flagship, and the smallest model statistically tied with it becomes the workhorse. Two models are tied with the flagship, and the smaller of them, Qwen3-8B, is therefore the workhorse. It trails the flagship by 0.0102 macro-F1 with a 95% confidence interval of $[-0.0225, +0.0021]$ that contains zero. One anomaly stands out: Qwen3-14B trails its own 8B sibling by 0.0056 macro-F1 and is *not* tied with the flagship. Within this family, scale did not buy identification quality under the shared recipe. Per-class results (Figure 2; full table in Appendix B) show the same two classes limiting every model. These are attack1, the uniform multiplicative reduction that most resembles a frugal consumer, and attack12, the consumption-profile substitution, with per-class F1 as low as 0.78 and 0.81 for the workhorse against a near-perfect 1.00 on attack0 and attack13. A rank sweep around the shared recipe (Appendix A) finds $r = 8$ statistically tied with the default and $r = 32$ significantly worse, so the screening was not capacity-limited and the single-recipe design was fair.

The classical rows of Table III, however, constrain how these gains should be read. The random forest reaches 0.9122 macro-F1 on the identical split, from the same input, at effectively zero inference cost; CatBoost and XGBoost land just below it (0.9093 and 0.9076), so the forest is the classical ceiling here and the tests that follow are against the strongest classical baseline. Paired on the same days, the workhorse's identification advantage over it is not statistically significant ($\Delta = +0.0107$, 95% CI $[-0.0055, +0.0265]$; McNemar $p = 0.204$); only the flagship separates ($\Delta = +0.0209$, CI $[+0.0051, +0.0368]$, $p = 0.0158$), and it does so against all three classical models ($p \leq 0.0048$ for CatBoost and XGBoost). What the workhorse does hold over the forest is binary detection: its specificity of 0.892 against the forest's 0.715 more than halves the false-alarm rate (McNemar $p < 10^{-5}$),

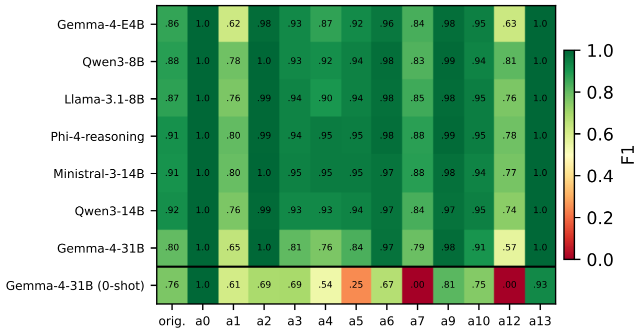

Fig. 2. Per-class F1 for the seven fine-tuned models, ordered by parameter count, with the best zero-shot model (Gemma-4-31B) as the bottom row for contrast. attack1 and attack12 limit every model.

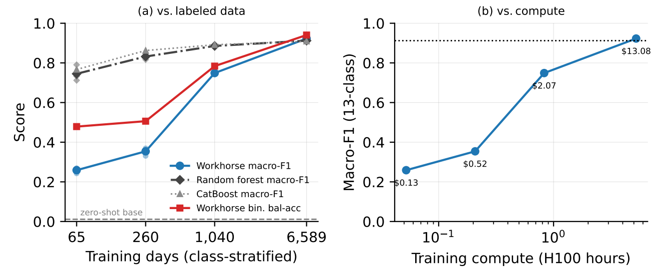

Fig. 3. The data-efficiency curves of the workhorse and the classical baselines trained on the identical subsets: (a) macro-F1 against training-set size, with the workhorse's binary balanced accuracy; (b) workhorse macro-F1 against training compute (the classical models train in CPU-minutes at every size). Per-seed points shown at N=65 and N=260; XGBoost (slightly below the forest throughout) is omitted for legibility and tabulated in Appendix D. Every classical variant dominates until full data.

TABLE V

IDENTIFICATION AND DETECTION QUALITY AGAINST TRAINING-SET SIZE FOR THE WORKHORSE AND THE RANDOM FOREST, BOTH TRAINED ON THE IDENTICAL CLASS-STRATIFIED SUBSETS (SEED MEANS OVER THREE DATA SEEDS AT N=65 AND N=260). COST ASSUMES AN H100 AT $2.50/H; THE FOREST TRAINS IN CPU-MINUTES AT EVERY SIZE.

N (per class)	Macro-F1	RF macro-F1	Bin. bal-acc	AUC	Specificity	Train time	Cost
65 (5)	0.2588	**0.7451**	0.4787	0.8869	0.008	3.1 min	$0.13
260 (20)	0.3540	**0.8321**	0.5061	0.9112	0.049	12.5 min	$0.52
1,040 (80)	0.7488	**0.8855**	0.7838	0.9711	0.585	49.7 min	$2.07
6,589 (full)	**0.9229**	0.9122	**0.9406**	**0.9939**	**0.892**	5.23 h	$13.08

Seed-level macro-F1, workhorse, N=65: 0.2427/0.2687/0.2650; N=260: 0.3620/0.3690/0.3310. RF, N=65: 0.7905/0.7327/0.7120; N=260: 0.8473/0.8324/0.8167. CatBoost seed means: 0.7667/0.8624/0.8915; XGBoost: 0.6919/0.8380/0.8790 (per N, ascending). Bold marks the better identifier per row. AUC and specificity at $N \leq 260$ are workhorse seed means.

and every avoided false positive is an avoided site inspection. Accuracy alone therefore justifies fine-tuning an LLM only weakly, and the following subsections examine what else the model provides. In summary, we can conclude that fine-tuned open models at 8–14B parameters match the zero-shot proprietary frontier. The selection rule hands every subsequent experiment to the workhorse, Qwen3-8B, beginning with the question of what the result costs in labels and compute.

B. Label and Compute Efficiency (RQ2)

Matching the frontier is only useful if the labeling cost is payable, since annotated theft days are scarce at any utility. This experiment therefore prices the workhorse's result. We retrain it on stratified subsets of 5, 20, and 80 labeled days per class (three data seeds at the two smallest sizes) and compare against the full 6,589-day run. Table V and Figure 3 report the curve.

Each row of the data-curve table retrains the workhorse with the same recipe on fewer training days, so the rows differ only in the amount of labeled data. Growing the set from 65 to 260 days buys 0.095 macro-F1 (about 0.5 points per 1,000 additional labels at this scale); growing it from 260 to 1,040 buys 0.395 (again about 0.5 per 1,000, sustained over a far larger step); growing it from 1,040 to 6,589 buys only 0.174 (0.03 per 1,000). The marginal value of a label collapses after the third point, which is why we place the elbow between 20 and 80 labels per class: at 1,040 days the model already holds 0.7488, or 81% of the full-data 0.9229. The full run costs 5.23 GPU-hours, roughly $13 at assumed rates, and the 1,040-day point costs $2.07 (per-run costs in Appendix F). The binding budget is label acquisition rather than compute.

The random forest, trained on the identical subsets, supplies the classical reference at every size, and below full data the comparison is one-sided. The forest holds 0.7451 macro-F1 at 65 days, 0.8321 at 260, and 0.8855 at 1,040, against the workhorse's 0.2588, 0.3540, and 0.7488. It also reaches a usable decision boundary immediately (specificity 0.52–0.77 at $N \leq 260$, where the workhorse sits near zero and would flag nearly every consumer for inspection). The conclusion is not an artifact of the estimator: CatBoost tracks slightly above the forest below full data (0.7667/0.8624/0.8915 seed means) and XGBoost slightly below it, so every classical variant we tested dominates the workhorse until labels are plentiful. Only at 6,589 days does the workhorse overtake them, and even there the identification gap to the strongest classical model is not statistically significant, as Section IV-A established; the full-data edge that survives testing is binary specificity. A utility with few labeled days is therefore better served by the forest, and the case for fine-tuning rests on the full-data regime together with the capabilities the rest of this section establishes. However, the more mechanistically interesting finding is in the binary columns. At $N \leq 260$ the model is a near-constant *manipulated* predictor (specificity 0.008–0.049) even though its AUC has already reached 0.887–0.911. The ranking is learned almost immediately; the decision boundary is what takes thousands of examples. Appendix H measures the same ordering in calibration terms: the expected calibration error of the verbalizer score falls monotonically from 0.078 zero-shot to 0.006 at full data, so the score becomes a usable probability only where the decision boundary does. This mirrors the zero-shot observation of Section IV-A from the opposite direction. It also suggests a practical shortcut we did not pursue: a utility with few labels could combine the early-learned ranking with a threshold calibrated on a small benign sample instead of learning the full boundary.

Identification follows the same order, and earlier than expected. At 65 training days the model already names the correct attack on 35.5% of manipulated days (more than four times the 8.3% chance rate) while its detector is still a near-constant; at 260 days the rate reaches 44.7%. The model learns to name attacks before it learns to withhold a flag on benign days, presumably because the benign class is one of thirteen and calibrating against it requires benign contrast the small subsets barely contain. The output format is learned before either: every prediction at every training size parses correctly, with zero format violations even at 65 days. This ordering has a deployment reading. The capability the cascade of Section IV-G buys from the LLM, adjudicating and naming days a cheap screen has already flagged, is the one that emerges first, so even a label-poor utility could pair a classical screen with a lightly trained adjudicator. In other words, labeled data buys the output format first, the attack taxonomy second, and the decision threshold last. In summary, we can conclude that the low-label regime belongs to the classical model, and that the workhorse's frontier match is purchased with the full 6,589 labeled days and $13 of compute. However, every day evaluated so far exhibits an attack each detector saw in training, an assumption no deployment can rely on, and the next experiment removes it for both detectors.

C. Generalization to Unseen Attack Families (RQ3)

Having established that the recipe is affordable, we now remove its most optimistic assumption: that the attack catalog at training time is complete. Real manipulation strategies evolve, and a deployed detector will eventually face a manipulation it was never trained on. We simulate this with four rotations. In each rotation, one entire attack family of Table I is removed from the training data: zero-suppression (R-Z), multiplicative (R-M), threshold (R-T), or shape substitution (R-S). The workhorse is retrained on the remaining attacks and then evaluated only on the held-out attacks plus all 130 benign days, so every manipulated day it sees at evaluation belongs to a family it never saw a training example of. The textual descriptions of the held-out attacks remain in the prompt, so the rotations also measure whether a described-but-never-demonstrated attack can be recognized from its description alone. The random forest is retrained on the identical reduced training sets and evaluated on the identical days, so both detector families face the same question and can be compared cell by cell. Table VI reports the per-rotation aggregates for both, Table VII the workhorse's recall on each never-seen attack, Figure 4 the same per-attack view graphically, and Figure 5 the identification confusion on the held-out attacks. **[EDIT: Suggested figure: a small train/eval matrix diagram of the rotation design (four rows, shaded held-out cells) would make the setup graspable at a glance.]**

For the workhorse, detection transfers well. Ten of the twelve never-seen attack types are detected at a recall of 0.85 or higher, five of them at 1.000, and specificity on the benign days is preserved in every rotation (0.869–0.938), so the transfer is not bought with false alarms. The two failures are attack4, a low-severity reduction confined to a sub-window ($\bar{\rho} = 0.084$, recall 0.437), and attack10, the energy-neutral time reversal ($\bar{\rho} = 0.043$, recall 0.269). These two attacks come from different families, one multiplicative and one shape substitution, but they share a statistical property: both barely move the day's energy. Quietness alone does not force a miss, since attack9 is comparably quiet ($\bar{\rho} = 0.043$) yet detected at

TABLE VI

OUT-OF-CLASS BINARY DETECTION PER HELD-OUT FAMILY ROTATION, EVALUATED ON THE HELD-OUT ATTACKS PLUS ALL 130 BENIGN DAYS, FOR THE WORKHORSE AND THE RANDOM FOREST RETRAINED ON THE SAME REDUCED SETS. HELD-OUT ATTACKS: R-Z = 0, 2, 13; R-M = 1, 3, 4, 5; R-T = 6, 7; R-S = 9, 10, 12.

Rot.	Detector	n	Bal-acc	F1	AUC	Spec.	Recall
R-Z	Workhorse	520	0.9179	0.9562	1.0000	0.8846	0.9513
	Random forest	520	0.8603	0.9546	0.9918	0.7231	0.9974
R-M	Workhorse	635	0.8874	0.9137	0.9435	0.9154	0.8594
	Random forest	635	0.8529	0.8761	0.9192	0.9077	0.7980
R-T	Workhorse	365	0.9325	0.9630	0.9910	0.8692	0.9957
	Random forest	365	0.7973	0.8584	0.9203	0.7308	0.8638
R-S	Workhorse	520	0.8462	0.8497	0.9171	0.9385	0.7538
	Random forest	520	0.8346	0.9312	0.9379	0.7154	0.9538

TABLE VII

DETECTION RECALL ON EACH NEVER-SEEN ATTACK TYPE, WITH THE MEAN REALIZED ENERGY-REMOVAL PROXY $\bar{\rho}$. THE TWO FAILURES, MARKED *, ARE BOTH STATISTICALLY QUIET ATTACKS.

Rotation	Attack	n	Recall	$\bar{\rho}$
R-Z	attack0	130	1.000	1.000
	attack2	130	1.000	0.136
	attack13	130	0.854	0.501
R-M	attack1	126	0.968	0.521
	attack3	130	1.000	0.554
	attack4*	119	0.437	0.084
	attack5	130	1.000	0.500
R-T	attack6	130	0.992	0.372
	attack7	105	1.000	0.649
R-S	attack9	130	1.000	0.043
	attack10*	130	0.269	0.043
	attack12	130	0.992	0.618

*Statistically quiet: $\bar{\rho} < 0.09$ (attack10 is energy-neutral by construction). attack9 is comparably quiet yet fully detected; its flat profile is distinctive by shape.

1.000; its flattened profile is the most distinctive shape in the taxonomy. This suggests that the ability that carries over to unseen attacks is sensitivity to removed energy, supplemented by shape only where the shape is unmistakable, and that the magnitude of the theft, far more than its mechanism, sets the detection floor.

Identification behaves very differently. On the held-out attacks, the workhorse names the correct attack essentially never: multiclass accuracy is 0.000 in every rotation except R-Z, where attack2 is partially recovered from its prompt description (precision 1.000, recall 0.292). Figure 5 shows what the model answers instead, and the misattributions are highly structured. For nine of the twelve held-out attacks, at least two thirds of the predictions fall on one or two trained attacks whose transformation produces a similar profile. The zeroed day of attack0 is attributed to threshold clipping (attack6, 0.99), which at a cut-off near zero produces the same all-zero day. The flat profile of attack9 is attributed to clipping at the observed constant (attack6, 1.00). The uniform scale-down of attack1 is read as the substituted lower-consuming profile (attack12, 0.81), and when attack12 is the held-out attack the attribution reverses to attack1 (0.86); the per-sample and pattern-replacing reductions attack3 and attack5 also land on attack12 (0.78, 0.79). The three remaining rows are consistent with the detection results: attack4 and attack10, the two detection failures, send most of their mass to the original column (0.56, 0.73), and attack7 spreads over several reduction mechanisms. The model's stated reasoning shows the same

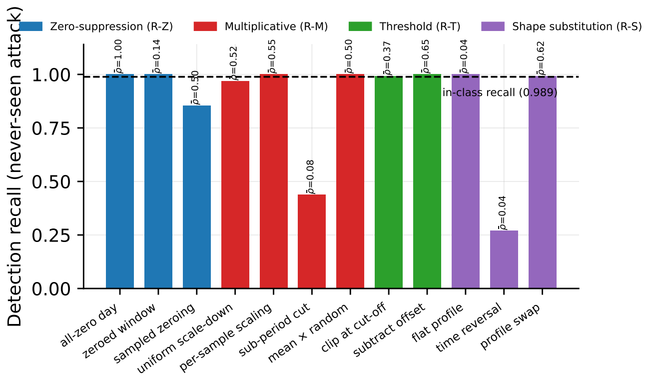

Fig. 4. Detection recall per never-seen attack, grouped by held-out rotation and annotated with the mean energy-removal proxy $\bar{\rho}$. The horizontal line is the in-class recall of the workhorse (0.989).

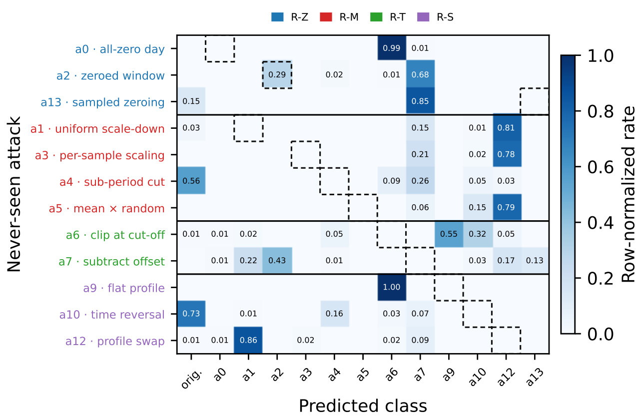

Fig. 5. Identification confusion for the never-seen attacks. Each row is a held-out attack, grouped and colored by rotation; each column is a predicted class, row-normalized over that attack's evaluation days. The dashed cell in each row marks the correct name. Prediction mass concentrates on trained attacks of similar mechanism, and the original column collects the detection misses of Table VII.

behavior on individual days. On a flat-profile day it writes that every interval reads exactly 27.0 Wh while the reference week fluctuates between roughly 20.6 and 41.3 Wh, and attributes the day to a clipping cut-off at 27.0 Wh: the description of the effect is accurate, and the inferred mechanism is wrong. The failure is therefore confined to the naming step. The model perceives and describes the distortion correctly, and lacking a training demonstration of the true class, it selects the trained mechanism most consistent with what it sees; the catalog text alone recovers the name only for attack2, whose on–off signature is the most distinctive description in the prompt. For a utility this failure mode is comparatively benign, since the investigator still receives a hypothesis of the right character, a scaled-down profile, zeroed intervals, or a flattened day, under a neighboring catalog name.

The random forest, facing the same rotations, fails in different places. At the aggregate level, the forest also transfers detection (recall 0.798–0.997). Its specificity sits at 0.715–0.731 on three of the four rotations, essentially its full-catalog level of 0.715, so its false-alarm rate is high with or without a complete catalog; the workhorse holds 0.869–0.938 in the same cells. On attack4, both fail and the forest fails harder (recall 0.227 against the workhorse's 0.437): where the energy signal is nearly absent, neither representation compensates. On attack10, the time reversal, the forest reaches 0.862 where the workhorse managed only 0.269, because its input features include the shape correlation that the reversal disturbs. On attack6, the threshold clipping, the roles invert: the workhorse holds 0.992 against the forest's 0.754. This suggests that the detection floor has two components, a shared one set by how little energy the attack moves, and a detector-specific one set by what each model's representation happens to encode. The pattern is robust across classical estimators: CatBoost and XGBoost reproduce all three signatures (attack4 at 0.286 and 0.235, attack10 at 0.946 and 0.846, attack6 at 0.715 for both; per-rotation aggregates in Appendix D). The two detector families therefore cover each other's blind spots, which is the property a cascade exploits.

One further regularity in the rotation table has operational value. The workhorse's specificity barely moves as the training catalog changes (0.869–0.938 across the four rotations, against 0.892 with the full catalog), so its false-alarm rate is anchored in its model of the benign consumer rather than in the attack catalog. This suggests that catalog updates, including retraining on newly discovered attack types, can be deployed without re-budgeting the inspection workload. In summary, we can conclude that detection survives novel attack families wherever meaningful energy is removed, for both detector families, and that the two fail on different attacks. The workhorse's two failures point at a single governing variable, which the next experiment measures directly.

TABLE VIII

DETECTION RECALL BY THEFT-SEVERITY BUCKET (ρ EDGES AT 10/30/60% OF ENERGY REMOVED) ON THE EVALUATION SLICE. THE SEVERITY GRADIENT IS CONFINED ALMOST ENTIRELY TO THE LIGHTEST BUCKET.

Model	<10% (n=138)	10–30% (n=195)	30–60% (n=529)	>60% (n=398)	Energy-neutral (n=260)
Qwen3-8B	0.957	0.990	0.998	1.000	0.969
Qwen3-14B	0.964	1.000	1.000	1.000	0.977
Phi-4-reasoning	**0.971**	1.000	0.998	1.000	0.977
Ministral-3-14B	**0.971**	1.000	0.998	1.000	0.965
Llama-3.1-8B	0.964	1.000	1.000	1.000	0.973
Gemma-4-E4B	0.884	0.995	1.000	1.000	0.969
Gemma-4-31B	0.768	0.964	0.996	1.000	0.954
Qwen3-8B (magnitude)	0.949	1.000	1.000	1.000	0.981

D. Severity-Stratified Failure Analysis (RQ4)

The out-of-class failures suggested that the amount of removed energy governs detectability. This experiment formalizes that hypothesis by stratifying every model's recall over the severity proxy ρ of Section III-E. Table VIII reports recall per severity bucket for the seven screened models and the magnitude variant of Section IV-F, with the energy-neutral pair (attack9 and attack10, which leave the day's total energy unchanged and alter only the profile) reported separately. Figure 6 plots the curves. The full table including multiclass accuracy and the low-data runs is in Appendix C.

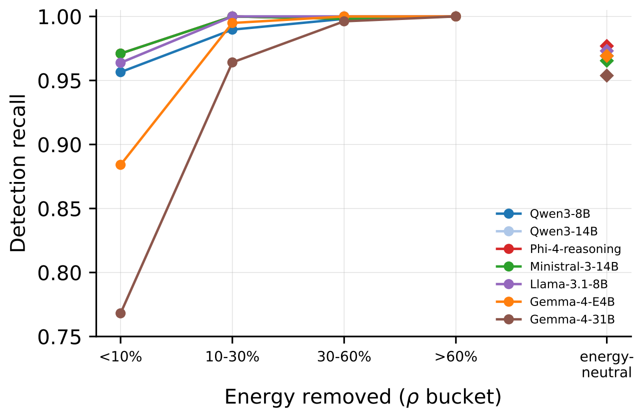

Fig. 6. Detection recall against theft severity for the screened models, with the energy-neutral pair (total energy unchanged, profile reshaped) as detached markers. Every model saturates above 60% energy removal; the differences live below 10%.

The stratification confirms the hypothesis. Every fine-tuned model reaches a recall of 1.000 in the >60% bucket, and the entire quality gradient between models is compressed into the <10% bucket, where the best models hold 0.971 and Gemma-4-31B drops to 0.768. That lightest bucket also carries a robustness reading. Benign week-to-week variation of ρ spans -0.197 to $+0.178$ (Section III-E), so a manipulation in the <10% bucket changes the day's energy by less than the consumer's ordinary fluctuation, yet five of the seven models detect these days at 0.957 or higher. This indicates that the detector discriminates manipulation structure from natural variation rather than thresholding removed energy, an observation that holds within the synthetic-attack setting in which it is measured. The energy-neutral pair is the unintuitive case in this ordering. An attack that removes no energy at all should, on this logic, be the hardest to detect, since the axis that governs every other failure offers nothing to see; yet the pair is detected at 0.954–0.981, above the <10% bucket. The explanation is that ρ measures only one kind of departure from normal consumption. The two energy-neutral mechanisms keep the day's total unchanged by distorting its profile more drastically than any reduction does: attack9 replaces the day with a constant at its own mean, and attack10 moves the evening's consumption into the morning, contradicting the reference week at almost every interval. A day scaled down by a few percent still resembles the consumer's ordinary days; a flat or time-reversed day resembles none of them. Judged by distance from the consumer's habitual shape rather than by energy removed, these attacks are among the easiest in the taxonomy, and the models detect them accordingly. The rotation experiment shows that this ease is learned rather than intrinsic: the time reversal that the workhorse detects at 0.938 when trained on it was detected at 0.269 when never demonstrated (Section IV-C). The intuition that energy-neutral theft is hard therefore holds only for mechanisms absent from training; once a single training pass has shown the model what a reshaped day looks like, wholesale reshaping becomes a strong signature. For the utility, the implication is direct.

TABLE IX
RATIONALE DISTILLATION ON THE WORKHORSE UNDER THE CATALOG-FREE PROMPT: CLASSIFICATION, EXPLANATION OVERLAP (ROUGE-L), AND PARAMETER-EXTRACTION FIDELITY. R0'S MACRO-F1 STEMS FROM A DIFFERENT RECIPE THAN THE SCREENING RUNS AND IS NOT COMPARABLE TO TABLE III.

Run	Train	Macro-F1	Bin. bal-acc	Spec.	ROUGE-L	Param. correct
R0 (label only)	6,589	**0.9444**	0.9384	0.885	—	—
R1 (rationale+label)	6,589	0.9285	0.9412	0.892	**0.454**	**0.798**

Parameter extraction covers the five attacks governed by a single recorded value (Section III-E; n=610).

Theft large enough to matter financially is detected essentially always, and the residual risk is confined to manipulations that remove less than a tenth of a day's energy, which are also the manipulations with the smallest per-day revenue impact. However, this finding sharpens rather than settles the objection raised at the end of Section IV-A. The random forest also detects consequential theft, as its rotation recalls showed, so if detection saturates for every competent model, the case for an LLM must rest on what it produces beyond the flag. In summary, we can conclude that detectors fail along the severity axis rather than by attack family, and we turn to the capability that classical baselines do not offer at all.

E. Rationale Distillation (RQ5)

A flagged day triggers a human decision, and the investigator's first question is *what happened*, since the flag itself already answered *whether*. This experiment asks whether the workhorse can be taught to answer that question, and what the answer costs in classification quality. We fine-tune two variants on the teacher rationales of Section III-E under a catalog-free prompt. The catalog is removed because the screening prompt describes all thirteen mechanisms, and a model could assemble a plausible-sounding explanation by paraphrasing the description of whichever class it predicts; without the catalog, the explanation must be grounded in the readings and in what fine-tuning placed in the weights. R0 is a label-only control; R1 trains on rationale plus label over the full 6,589 days and differs from R0 in nothing else, so any gap between them prices the rationale target itself. Table IX reports classification and explanation quality; the paired contrast, specified before the run, is stated in the text.

Three findings emerge. First, rationale supervision carries a small but real classification cost. Training on rationales costs -0.0159 macro-F1 against the label-only control (95% CI $[-0.0284, -0.0037]$; McNemar $p = 0.0126$ on 93 discordant days), a small but statistically significant identification penalty. Binary detection, however, is statistically unchanged (Δ balanced accuracy $+0.0029$, CI $[-0.0266, +0.0320]$), so the utility pays nothing in detection to gain an explanation for every flag. Second, the explanations survive quantitative checks. R1 reaches a ROUGE-L of 0.454 against the teacher and, on the stricter test, states the numerically correct attack parameter on 79.8% of the 610 days from attacks with a stated parameter. The rate tracks how directly the parameter is visible in the readings. The clipping cut-off of attack6, which appears literally as the day's maximum, is stated correctly on 98.5% of its days; the factor and window parameters of attack1, attack2, and attack4 follow at 82.5–88.5%; the subtracted offset of attack7, which must be inferred from the residual profile, drops to 37.1%. In other words, the model reliably reports what it can read and struggles with what it must reconstruct. Third, the student's fidelity traces the teacher's own difficulty. R1's overlap with the reference is 0.463 on days the teacher diagnosed unaided but 0.384 on the rationalization-fallback days, where the teacher first erred and was handed the label; the same ≈0.46-versus-0.38 split recurs in every full-data run we scored (Appendix E). The fallback marks, by construction, the days whose readings least determine the attack, so the student struggles where the teacher did. This suggests the rationale channel transmits not only the teacher's competence but the difficulty structure of the task, and it cautions against reading fidelity as a single number: the hard tail is where explanations degrade first. One scope limit applies: R0's headline macro-F1 of 0.9444, the highest in this paper, comes from a different recipe than the screening runs and therefore does not enter the model comparison of Section IV-A.

Overlap and parameter fidelity establish that the rationale is grounded in the correct numeric evidence, but neither instrument judges whether the explanation is behaviorally and factually correct as a piece of reasoning. We therefore run an independent LLM-as-judge evaluation on the workhorse's rationale generations (the R1 run of Table IX). The judge is GPT-4o at temperature 0, deliberately drawn from a different model family than the teacher, since Gemini 3 Pro authored the rationales the workhorse was distilled from and a same-family judge would introduce self-preference bias between the answer key and the grader. Each generation is scored against a per-row reference built deterministically from the day's own recorded ground-truth parameters rather than from the teacher's own rationale text, which avoids a second circularity: scoring the student against the text it was trained to imitate would measure imitation fidelity rather than correctness, and would silently reward any error the teacher's own rationale contained. Seven attack classes with no single per-day parameter reuse a hand-authored, attack-type-level reference; the five parameterized attacks instead receive a template reworded per row to state that day's actual ground-truth value, so the reference is exact rather than merely representative. The judge rates each explanation 1–5 on four axes, with reasoning required before each score: behavioral correctness (BC, whether the explanation correctly describes how the attack modifies the consumption pattern), completeness (COM, whether the explanation covers every key aspect of the attack rather than omitting an important detail), faithfulness to the data (FTD, whether its claims are consistent with the actual attack rather than hallucinated), and coherence (COH, clarity and internal consistency).

A total of 1,461 of the 1,520 manipulated evaluation days completed the judging pass. Table X reports the results overall and per attack class, alongside a locally computed classification-correctness side channel: the workhorse's free-text rationale ends with a concluding attack label, which we extract by regex and compare to ground truth over each class's full set of manipulated evaluation days. Overall, the workhorse reaches a mean judge score of 4.29 (BC 4.25, COM 4.10, FTD 4.24, COH 4.56) with 91.1% classification accuracy over the manipulated evaluation days. Judge score and classification correctness are correlated but not redundant. Conditioning the judged rows on whether the rationale's self-declared label matches ground truth, mean judge quality is 4.37 on correctly classified days (n=1,345) against 3.33 on misclassified ones (n=116), so getting the label right and describing the attack's behavior correctly are related but separable competencies. This separation shows up concretely between attack3 and attack5, two mechanisms from the same multiplicative/randomized family whose per-interval randomization leaves neither with a single clean scaling factor to identify, unlike attack1's uniform multiplier or attack2's contiguous window; misclassified days between the two can still score 4/4/4/4. An explanation names the mechanism; an inspector planning recovery also needs its size, which is the final capability we test. In summary, we can conclude that mechanism explanations cost 1.6 macro-F1 points and no detection while scoring 4.29 of 5 under an independent cross-family judge, and we next extend the same output schema from naming the theft to quantifying it.

TABLE X
LLM-AS-JUDGE SCORES (GPT-4O, TEMPERATURE 0, 1–5 SCALE) FOR THE WORKHORSE'S RATIONALE GENERATIONS; n COUNTS THE JUDGED ROWS PER CLASS (1,461 IN TOTAL). CLASSIFICATION ACCURACY IS EXTRACTED FROM THE RATIONALE'S SELF-DECLARED LABEL AT NO ADDITIONAL JUDGE COST AND IS COMPUTED OVER EACH CLASS'S FULL SET OF MANIPULATED EVALUATION DAYS.

Attack	BC	COM	FTD	COH	Mean	n	Cls. acc.
Overall	4.25	4.10	4.24	4.56	**4.29**	1,461	91.1%
attack0	5.00	5.00	5.00	5.00	5.00	130	100.0%
attack1	3.77	3.68	3.75	4.44	3.91	117	84.9%
attack2	4.53	4.57	4.57	4.76	4.61	127	99.2%
attack3	4.73	4.28	4.72	4.83	4.64	126	92.3%
attack4	3.25	3.40	3.28	4.12	3.51	106	92.4%
attack5	3.86	3.78	3.65	4.01	3.82	126	93.9%
attack6	3.79	3.77	3.71	4.38	3.91	126	97.7%
attack7	3.15	3.29	3.09	3.92	3.36	96	83.8%
attack9	4.97	4.94	4.96	4.98	4.96	129	98.5%
attack10	4.54	4.31	4.54	4.68	4.52	124	92.3%
attack12	3.99	3.84	4.13	4.50	4.12	127	56.9%
attack13	4.98	4.02	4.92	4.86	4.69	127	100.0%

F. Quantifying Theft Magnitude (RQ5)

Extending explanation to quantification, we append a magnitude block to the structured output: the estimated fraction of energy removed and the manipulated time window. The workhorse is fine-tuned to emit it alongside the classification. The ground truth is exact rather than estimated: the attacks are synthetic, so every attacked day's pre-attack original is on record, and the true removed fraction follows as $1 - E_{\text{attacked}}/E_{\text{pre-attack}}$. Two questions matter. The first is whether the estimates are genuine per-day measurements rather than class-conditional averages; the second is whether carrying the extra target degrades classification. Table XI answers the first; the interference check, specified before the run, answers the second.

The answer to the first question is yes. The model attains an MAE of 0.0412 on the removed fraction over the 1,260

TABLE XI
EXPLICIT MAGNITUDE QUANTIFICATION BY THE WORKHORSE: HEADLINE ERROR, CALIBRATION, AND LOCALIZATION (TOP), AGAINST PER-DAY CONTROLS (BOTTOM). MAE IS ON THE ENERGY-REDUCTION FRACTION $\rho \in [0, 1]$ AGAINST THE PRE-ATTACK DAY, $n = 1{,}260$ MAGNITUDE-BEARING DAYS.

Quantity	Value
MAE (95% CI)	$0.0412\ [0.0373, 0.0456]$
Pearson / Spearman	0.978 / 0.956
Severity-bucket accuracy	0.825
Window coverage / mean IoU	0.956 / 0.874
Format-violation rate	0.000

Control	MAE	RMSE	95% CI
Workhorse (LLM)	**0.0412**	**0.0598**	$[0.0373, 0.0456]$
Class-prior (predicted class)	0.0737	0.1275	$[0.0701, 0.0776]$
Class-prior (gold class)	0.0718	0.1242	$[0.0682, 0.0755]$
Ridge on readings	0.1446	0.1712	$[0.1387, 0.1513]$
GBM on readings	0.0319	0.0557	$[0.0266, 0.0381]$

magnitude-bearing days, with a Spearman correlation of 0.956. It beats the class-prior control (the strategy of looking up the average severity of the predicted attack class) by 0.0326 MAE with a confidence interval excluding zero. The same holds against the prior computed from the gold class, so the advantage is not explained by classification accuracy. Localization is similarly strong (window IoU 0.874, boundary error under one 15-minute interval; per-attack breakdown in Appendix G). The model correctly declines to claim magnitude on the energy-neutral attacks (mean claimed ρ of 0.012 against a true 0.000), and the output format is never violated. However, a gradient-boosted regressor reading the same raw values attains an MAE of 0.0319, so a dedicated regressor remains the modestly better tool for magnitude in isolation. The interference check, specified before the run, answers the second question: adding the magnitude target does not degrade classification relative to the plain workhorse (Δ macro-F1 $+0.0057$, CI $[-0.0043, +0.0157]$; McNemar $p = 0.808$ binary, 0.245 multiclass). Set against Section IV-E, the numerically harder auxiliary task (per-day magnitude regression) costs classification nothing, while the linguistically heavier one (free-text rationales) costs a small but significant 1.6 macro-F1 points. This suggests that the price of an auxiliary target tracks the token mass it adds to the training signal rather than the difficulty of the task, since the magnitude block is a handful of tokens and a rationale is a paragraph, and the training losses point the same way (0.014 label-only against 0.42 with rationales). We offer this as an observation from two data points only. In other words, the model's value lies in the integrated output: a single structured answer that reports the detection, the mechanism, the estimated size, and the manipulated window, at no measured cost to detection. In summary, we can conclude that the magnitude estimates are accurate at the day level and close to a dedicated reading-level regressor, and the deployment analysis that follows weighs the whole package against its alternatives.

G. Deployment Cost and Recommended Workflow (RQ6)

The preceding sections measured the fine-tuned detector's accuracy, its robustness to unseen attacks, and the outputs it attaches to each flag. This final analysis turns those results into the deployment decision a utility actually faces, and closes with the workflow they recommend. The decision is between two cost structures rather than two prices: an API detector is billed per query-day indefinitely ($5 per million tokens at $\sim$6,500 tokens per query-day, or $32.50 per 1,000 query-days), while a local detector is an initial investment, the GPU and a one-off $13.08 fine-tune, after which a query-day costs effectively nothing because the hardware is owned and on premises. Table XII reports every operating point; the dollar column states rental-equivalent rates at an amortized $2.50/h so all points share one axis, and the throughput column is the binding quantity under ownership.

TABLE XII
THE DEPLOYMENT FRONTIER AT ASSUMED PRICES: IDENTIFICATION QUALITY AGAINST INFERENCE COST PER 1,000 QUERY-DAYS. PARETO-OPTIMAL POINTS ARE BOLDED; THE CASCADE USES THE RANDOM FOREST TO SCREEN AND THE WORKHORSE TO ADJUDICATE FLAGGED DAYS. DOLLAR FIGURES ARE RENTAL-EQUIVALENT (AMORTIZED $2.50/H); UNDER ON-PREMISES OWNERSHIP THE BINDING COLUMN IS THROUGHPUT (S/DAY).

Operating point	Kind	Macro-F1	$/1k days	s/day
Random forest	classical, CPU	**0.9122**	**0.01**	—
Qwen3-8B (workhorse)	local LLM	**0.9229**	**7.48**	10.8
Ministral-3-14B	local LLM	0.9298	9.07	13.1
Phi-4-reasoning (flagship)	local LLM	**0.9331**	**9.07**	13.1
Gemma-4-31B	local LLM	0.8517	33.89	48.8
Gemini 3.1 Pro	API, zero-shot	0.929	32.50	—
Cascade (RF $\rightarrow$ workhorse)	hybrid	0.9114	6.92	—

Flagship throughput estimated from the same-size Ministral-3-14B; all costs assumption-driven.

On accuracy the ordering is unchanged: the Pareto front runs random forest $\rightarrow$ workhorse $\rightarrow$ flagship, and both the API reference and the 31B model sit strictly inside it, the API offering 0.006 more macro-F1 than the workhorse at 4.3 times its rental-equivalent rate. The ownership view turns this comparison into payback arithmetic. Every query-day adjudicated locally rather than through the API saves the full $0.0325, so the $13.08 fine-tune is repaid after roughly 400 query-days (523 under the rental-equivalent accounting), and a GPU purchased at an assumed $30,000 is repaid after roughly 900,000: a fleet of 2,500 meters screened daily reaches that point in about a year, and a 25,000-meter fleet in about five weeks. The fine-tune does not require owning the training card, since its 5.23 GPU-hours can be rented for $13.08, and the workhorse's 8B weights (roughly 16 GB at 16-bit) fit inference cards well below the H100 used for our timings, so the assumed $30,000 is an upper bound on the entry price. What remains recurring is the label bill of Section IV-B, paid per catalog update (6,589 labeled days for the numbers here, with entry at 1,040 labels and 81% of the quality), and electricity, which at $\sim$11 GPU-seconds per query-day is negligible against every figure above. Consumption data also never leaves the utility's premises, one of the motivations stated in Section III. The random forest consumes the identical labeled set, so at full data the comparison is label-neutral; below full data, Section IV-B showed the forest degrading far more gracefully (0.745 macro-F1 at 65 days against the workhorse's 0.259), so there the classical screen remains the better detector.

These results resolve the question Section IV-A left open, what justifies an LLM against a near-free classical model of comparable accuracy, in favor of pairing the two. The recommended workflow is:

1) **Screen:** the random forest scores every meter-day on CPU at \$0.01 per thousand. It is the right first stage because it is the strongest detector in the low-label regime (Section IV-B) and its detection transfers to never-seen attack families (Section IV-C).
2) **Adjudicate:** days the screen flags are re-decided by the fine-tuned workhorse, whose specificity of 0.892 against the forest's 0.715 more than halves the false alarms that survive to inspection, and which attaches to every confirmed flag the attack identification, the mechanism explanation (Section IV-E), and the magnitude estimate with its manipulation window (Section IV-F).
3) **Inspect:** the investigator receives only explained and quantified findings, each carrying a named mechanism, an estimated energy loss, and the manipulated window as the starting point of the case file.

On the evaluation slice this cascade retains 0.9114 macro-F1. Under the ownership view its role is to size the investment. One card adjudicates about 8,000 query-days per day at the workhorse's 10.8 s each, and the free screen in front of it reduces the LLM's load to the flagged fraction of the fleet, so the number of cards to buy follows from the expected flag rate rather than from the fleet size. The complementary failure profiles of Section IV-C are a further reason to pair the detectors rather than choose between them, since each covers attacks the other misses. Because the workhorse's false-alarm rate is anchored in its model of the benign consumer rather than in the attack catalog (Section IV-C), newly discovered attack types can be retrained into the adjudication stage, at \$13 of rented compute per update, without re-budgeting the inspection stage. However, the cascade's flag rate on this theft-heavy evaluation slice (0.925) is far above what a deployed screening population would produce, so sizing should use the utility's own flag history. In summary, we recommend the cascade as the deployment default: the classical screen remains the free first stage, the fine-tuned workhorse converts each flag into an explained and quantified finding, and the up-front cost, one or a few inference cards and a \$13.08 fine-tune, is sized by the flag rate rather than the fleet.

H. Threats to Validity and Limitations

While these experiments support the conclusions above, certain limitations should be acknowledged. Firstly, all attacks are synthetic manipulations of one real consumption dataset. The taxonomy is standard in the non-technical-loss literature, but real theft need not respect it, and our own out-of-class results show detection degrading where an attack departs from the energy-removal pattern the taxonomy embodies. The one formatting cue the injection could create, floating-point values off the meter's quantization grid, is removed at preprocessing by re-quantizing every manipulated reading to the meter's native step (Section III-B). Secondly, the classical comparison is bounded by its coverage: the tree-family estimators are fully matched across the full-data, data-size, and rotation grids, but the logistic regression and the local outlier factor are recorded as balanced accuracy only and provide binary-detection context rather than paired comparisons. Thirdly, the rationale evaluation is bounded by its instruments: ROUGE-L rewards overlap rather than truth, parameter fidelity is defined only for the attacks governed by a single recorded value, and the LLM-judge scores, though drawn from a model family disjoint from the teacher's, remain model judgments rather than human ones. Similarly instrument-bound, the verbalizer score's calibration (Appendix H) is measured against the evaluation slice's 92.1% manipulated prior, and reading the score as a probability at fleet prevalence would first require recalibrating it to the operating prior. Fourthly, every deployment cost rests on assumed prices (a \$2.50/h rental-equivalent, blended API tokens, and a \$30,000 GPU purchase for the payback arithmetic). The frontier's ordering is robust to price changes, but the absolute dollar figures and payback horizons are estimates. Finally, the detection floor identified in Sections IV-C and IV-D, statistically quiet manipulation, is a structural limitation of consumption-based detection shared by every detector we tested. Adversaries aware of it could deliberately operate below the 10% band, at proportionally reduced benefit. None of these threatens the direction of the results, but each would tighten a bound this paper leaves loose.

V. Conclusion

This paper asked whether the detection quality that a proprietary zero-shot language model established for energy theft can be reached by open-weight models a utility can fine-tune and run on its own hardware. The screening answers yes: the best of seven fine-tuned models reaches a macro-F1 of 0.9331 against the proprietary frontier's 0.929, and the 8B workhorse that carries the remainder of the study is statistically indistinguishable from that flagship. The screening also sharpened the question rather than settling it. A random forest refit on the identical split reaches 0.9122, so accuracy alone justifies the language model only weakly, and its durable advantages are the halved false-alarm rate (specificity 0.892 against 0.715) and the content it attaches to every flag.

The remaining experiments priced and stress-tested that position. Eighty labeled days per class recover 81% of full-data identification, and the full fine-tune costs 5.23 GPU-hours, roughly \$13 at assumed rates, so the binding budget is label acquisition rather than compute; until labels number in the thousands, the classical model dominates outright. Detection survives attacks the model never trained on, with ten of twelve held-out mechanisms detected at a recall of 0.85 or higher, while identification collapses on them, typically misattributing an unseen attack to the trained mechanism whose transformation is most similar. Failure concentrates along severity rather than attack family: every fine-tuned model detects essentially all theft above 60% of a day's energy, and the entire quality gradient sits below 10% removal, where the best models still hold 0.971. The explanation channel costs 1.6 macro-F1 points of identification and no detection, states the governing attack parameter correctly on 79.8% of parameterized days, and scores 4.29 of 5 under a judge drawn from a different

TABLE XIII
LoRA-RANK SWEEP AROUND THE SCREENING DEFAULT ON THE WORKHORSE. PAIRED STATISTICS ARE AGAINST $r = 16$.

Rank	Macro-F1	Bin. bal-acc	AUC	Spec.	Δ vs. r=16	McNemar p
$r = 8$	0.9140	0.9454	0.9943	0.900	−0.0089	0.0881
$r = 16$ (default)	**0.9229**	0.9406	0.9939	0.892	—	—
$r = 32$	0.9073	0.9444	0.9930	0.900	−0.0156	0.0158

model family than its teacher. The same output extends to magnitude: the stated stolen fraction lands within 0.0412 of the exact pre-attack value on average, with the manipulated window localized to under one 15-minute interval.

For deployment, the economics divide by cost structure rather than by accuracy. An API detector is billed indefinitely, whereas a locally fine-tuned detector is an initial investment with near-zero marginal inference on premises, repaid after roughly 400 query-days of equivalent API load. The workflow the results recommend is therefore a cascade: the random forest screens every meter-day at negligible cost, the workhorse adjudicates the flagged residue and attaches mechanism, magnitude, and window to each case, and inspectors receive a ranked, explained shortlist sized to the inspection budget.

Future work follows the bounds the limitations leave open: a time-forward evaluation protocol, validation on field-confirmed theft rather than synthetic manipulation, human assessment of the explanations alongside the model judge, and recalibration of the verbalizer score to fleet prevalence. However, the structural constraint is the quiet-theft floor: manipulation that removes less than a tenth of a day's energy remains hard for every detector we tested, ours and classical alike, and it bounds what consumption-based detection alone can deliver.

Appendix A
Recipe Validation: LoRA Rank

The screening of Section IV-A applied one shared recipe to all seven models, which is only fair if that recipe was not capacity-limiting. We validate this on the workhorse by sweeping the LoRA rank one step below and above the default. Table XIII shows that halving the rank is statistically indistinguishable from the default (Δ macro-F1 -0.0089, 95% CI $[-0.0197, +0.0015]$, McNemar $p = 0.0881$). Doubling it is significantly *worse* ($\Delta = -0.0156$, CI $[-0.0277, -0.0037]$, $p = 0.0158$). The results suggest that the shared recipe sat at or slightly above the capacity the task requires, so no screened model was starved by it.

Appendix B
Per-Class Results

Table XIV reports per-class F1 for all seven fine-tuned models with the best zero-shot model as the contrast row; Table XV the per-class F1 of every zero-shot base; Figure 7 the flagship's confusion matrix. Across every fine-tuned model, attack1 (uniform multiplicative reduction) and attack12 (profile substitution) are the two limiting classes, while attack0 and attack13 are essentially solved.

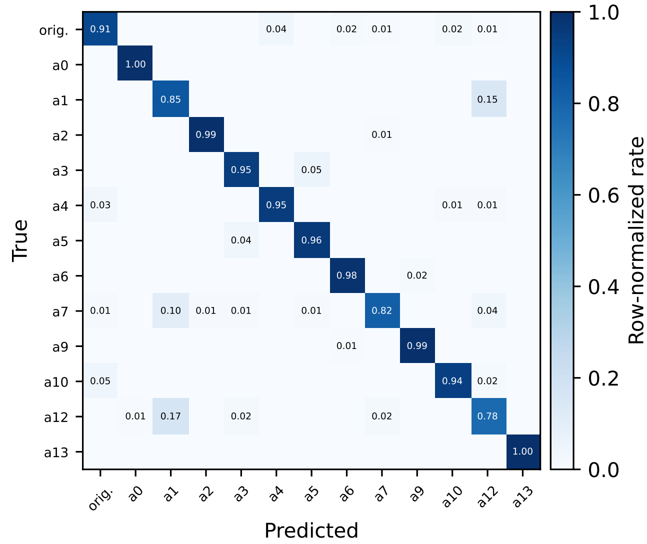

Fig. 7. Row-normalized confusion matrix of the flagship (Phi-4-reasoning) on the evaluation slice. The dominant confusions run between attack1 and the benign class, and between attack12 and the benign class.

Appendix C
Severity Stratification, All Runs

Table XVI extends Table VIII to every prediction set in the paper, reporting detection recall and multiclass accuracy per severity bucket. One pattern is worth noting beyond the main text. The low-data runs ($N \leq 260$) show high recall in every bucket only because they flag nearly everything; compare their specificity in Table V. Their multiclass accuracy collapses instead.

Appendix D
Classical Grid, All Estimators

Tables XVII and XVIII report the full classical grid (random forest, CatBoost, and XGBoost) on the matched data-curve subsets and rotation retrainings. Three observations carry to the main text. First, the ordering below full data is CatBoost $>$ random forest $>$ XGBoost, and every estimator dominates the workhorse until full data. Second, at full data the ordering inverts and the random forest is the classical ceiling (0.9122), which is why the paired tests of Section IV-A target it. Thirdly, the rotation signatures (attack4 missed, attack10 caught, attack6 weakened) appear in all three estimators, so the two-component floor of Section IV-C describes the whole feature family rather than one model.

Appendix E
Explanation Quality Across All Models

Table XIX reports ROUGE-L, parameter-mention rate, and parameter-correctness for every scored run. The screening models, trained with the attack catalog in the prompt, produce rationales of the same ROUGE-L band (0.43–0.46) as the dedicated R1 distillation run. This suggests that with catalog-prompt SFT, explanation ability comes essentially free; the catalog-free R1 run matters because it internalizes that ability

TABLE XIV
PER-CLASS F1 ON THE EVALUATION SLICE FOR THE SEVEN FINE-TUNED MODELS, WITH THE BEST ZERO-SHOT MODEL (GEMMA-4-31B) AS THE BOTTOM ROW. BEST VALUE PER CLASS COLUMN IS BOLDED WITHIN THE FINE-TUNED ROWS.

Model	orig.	a0	a1	a2	a3	a4	a5	a6	a7	a9	a10	a12	a13	Macro
Phi-4-reasoning	.911	**.996**	**.805**	.992	.939	**.954**	.951	**.981**	.878	**.989**	**.953**	.783	**1.00**	**.933**
Ministral-3-14B	.909	**.996**	.795	.996	**.945**	.953	**.955**	.969	**.880**	.977	.938	.773	**1.00**	.930
Qwen3-8B	.882	**.996**	.783	**1.00**	.929	.917	.943	**.981**	.826	.988	.941	**.811**	**1.00**	.923
Llama-3.1-8B	.873	**.996**	.763	.992	.941	.899	.943	.977	.852	.985	**.953**	.762	**1.00**	.918
Qwen3-14B	**.915**	**.996**	.765	.988	.929	.928	.940	.969	.836	.973	.946	.739	**1.00**	.917
Gemma-4-E4B	.865	**.996**	.622	.984	.927	.871	.924	.965	.836	.981	.949	.626	.996	.888
Gemma-4-31B	.804	**.996**	.647	.996	.808	.765	.835	.973	.791	.985	.906	.566	**1.00**	.852
Gemma-4-31B (zero-shot)	.762	.996	.613	.689	.694	.539	.248	.670	.000	.815	.750	.000	.929	.593

TABLE XV
PER-CLASS F1 OF EVERY ZERO-SHOT BASE MODEL ON THE EVALUATION SLICE.

Base model	orig.	a0	a1	a2	a3	a4	a5	a6	a7	a9	a10	a12	a13	Macro
Qwen3-8B	.146	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.011
Qwen3-14B	.147	.206	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.027
Phi-4-reasoning	.146	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.000	.011
Ministral-3-14B	.217	.869	.390	.382	.030	.017	.000	.000	.000	.766	.015	.000	.478	.243
Llama-3.1-8B	.040	.743	.152	.405	.000	.180	.292	.124	.000	.733	.015	.000	.043	.210
Gemma-4-E4B	.027	.594	.298	.616	.000	.377	.028	.015	.000	.780	.015	.000	.868	.278
Gemma-4-31B	.762	.996	.613	.689	.694	.539	.248	.670	.000	.815	.750	.000	.929	.593

into the weights. Every run also reproduces the fidelity split of Section IV-E: ROUGE-L of 0.44–0.47 on days the teacher diagnosed unaided against 0.34–0.39 on the rationalization-fallback days, so the difficulty signature is a property of the task rather than of one student.

APPENDIX F
TRAINING COST

Table XX records the training cost of every run, from the per-run metadata. All runs execute on a single H100 80 GB; the amortized figures of Section IV-G derive from these wall-clock times at $2.50/h. The 4-bit 31B model is the outlier at 22.4 hours, since quantized training trades memory for time.

APPENDIX G
PER-ATTACK MAGNITUDE ERROR

Table XXI decomposes the magnitude MAE of Section IV-F per attack. The total bypass (attack0), whose true ρ is identically 1, is estimated exactly. The hardest attacks for magnitude are those whose reduction is confined to a sub-window of the day or produced by clipping (attack2, attack4, attack6). On the energy-neutral pair the model claims a mean ρ of 0.012 against a true 0.000; it correctly declines to assert magnitude where there is none to measure. On the 130 benign days it never emits a magnitude block for a day it classifies as original, and its 18 false positives carry a mean claimed ρ of 0.067.

APPENDIX H
CALIBRATION OF THE VERBALIZER SCORE

The verbalizer score is used throughout as a ranking statistic, and Figure 8 asks the stricter question of whether it can be read as a probability. We bin the score into ten equal-width bins on the evaluation slice and plot the empirical manipulated rate per bin against the mean score in that bin, with the expected calibration error (ECE) summarizing the gap; multi-seed stages report the seed mean. Calibration arrives with training data, and it arrives after discrimination. The zero-shot base already ranks (AUC 0.907) but its score is saturated near 1.0 on every day (ECE 0.078), so it carries no probability content. At 65 and 260 labeled days the ranking is essentially in place (mean AUC 0.887 and 0.911) while the ECE remains at 0.070 and 0.044, and the full-data run closes it to 0.006 with a Brier score of 0.014, at which point the mean claimed probability (0.923) matches the evaluation prior (0.921). In other words, the ordering of Section IV-B, ranking first and decision behavior last, holds for calibration as well, as a measurement rather than an inference. However, the evaluation prior is 92.1% manipulated, so calibration is measured against that skewed base rate; a deployment at fleet prevalence would recalibrate the score to its operating prior before reading it as a probability.

APPENDIX I
REMARK ON THE LEGACY 32B PILOT

An earlier pilot fine-tuned Qwen3-32B with 4-bit QLoRA under the project's original data regime. It was evaluated on a 130-day class-stratified subset of a different holdout, with decision thresholds calibrated on a validation split. Its results (binary balanced accuracy up to 0.977 at a specificity of 1.0 under verbalizer thresholding; multiclass macro-F1 0.833) are not comparable to any table in this paper: different evaluation set, different size, and threshold tuning that the journal-round

TABLE XVI
DETECTION RECALL / MULTICLASS ACCURACY PER SEVERITY BUCKET FOR EVERY PREDICTION SET (EVALUATION SLICE; BUCKET n = 138/195/529/398, ENERGY-NEUTRAL n = 260).

Run	<10%	10–30%	30–60%	>60%	Energy-neutral
Qwen3-8B (workhorse)	0.957 / 0.942	0.990 / 0.923	0.998 / 0.904	1.000 / 0.940	0.969 / 0.950
Qwen3-14B	0.964 / 0.913	1.000 / 0.928	1.000 / 0.898	1.000 / 0.912	0.977 / 0.965
Phi-4-reasoning (flagship)	0.971 / 0.957	1.000 / 0.944	0.998 / 0.923	1.000 / 0.925	0.977 / 0.965
Ministral-3-14B	0.971 / 0.957	1.000 / 0.954	0.998 / 0.906	1.000 / 0.930	0.965 / 0.954
Llama-3.1-8B	0.964 / 0.949	1.000 / 0.949	1.000 / 0.906	1.000 / 0.910	0.973 / 0.962
Gemma-4-E4B	0.884 / 0.848	0.995 / 0.923	1.000 / 0.856	1.000 / 0.872	0.969 / 0.962
Gemma-4-31B	0.768 / 0.761	0.964 / 0.856	0.996 / 0.828	1.000 / 0.837	0.954 / 0.938
Qwen3-8B (magnitude)	0.949 / 0.920	1.000 / 0.949	1.000 / 0.915	1.000 / 0.947	0.981 / 0.962
$r = 8$	0.949 / 0.928	1.000 / 0.918	0.998 / 0.883	1.000 / 0.930	0.977 / 0.954
$r = 32$	0.928 / 0.906	0.990 / 0.923	0.996 / 0.881	1.000 / 0.910	0.989 / 0.958
R0 (label only)	0.957 / 0.949	1.000 / 0.954	1.000 / 0.936	1.000 / 0.957	0.977 / 0.969
R1 (rationale+label)	0.942 / 0.935	1.000 / 0.949	0.996 / 0.913	1.000 / 0.935	0.981 / 0.954
N=1,040	0.928 / 0.804	0.990 / 0.856	1.000 / 0.667	0.995 / 0.754	0.954 / 0.896
N=260 (s42)	0.993 / 0.399	0.985 / 0.523	0.945 / 0.250	0.995 / 0.412	0.996 / 0.946
N=260 (s43)	0.978 / 0.370	0.990 / 0.518	0.896 / 0.268	0.980 / 0.410	0.973 / 0.858
N=260 (s44)	0.993 / 0.478	0.990 / 0.600	0.928 / 0.174	0.982 / 0.417	0.969 / 0.846
N=65 (s42)	0.978 / 0.203	0.969 / 0.380	0.890 / 0.113	0.970 / 0.427	0.989 / 0.800
N=65 (s43)	0.935 / 0.174	0.949 / 0.339	0.964 / 0.187	0.982 / 0.455	0.965 / 0.658
N=65 (s44)	0.935 / 0.138	0.923 / 0.313	0.915 / 0.176	0.965 / 0.455	0.962 / 0.712

TABLE XVII
CLASSICAL DATA-CURVE CELLS: MACRO-F1 PER TRAINING-SET SIZE AND DATA SEED (MATCHED SUBSETS).

N	Seed	Random forest	CatBoost	XGBoost
65	42	0.7905	**0.8126**	0.7180
	43	0.7327	**0.7587**	0.6459
	44	0.7120	**0.7288**	0.7118
260	42	0.8473	**0.8706**	0.8417
	43	0.8324	**0.8675**	0.8245
	44	0.8167	**0.8491**	0.8477
1,040	42	0.8855	**0.8915**	0.8790
6,589 (full)	—	**0.9122**	0.9093	0.9076

runs deliberately forgo. We cite them only to acknowledge the pilot's existence. Re-scoring this checkpoint on the present benchmark would make it comparable; we leave that outside the present scope.

TABLE XVIII
CLASSICAL BASELINES UNDER THE ROTATIONS: OUT-OF-CLASS BINARY DETECTION PER HELD-OUT FAMILY (EVALUATED ON HELD-OUT ATTACKS PLUS ALL 130 BENIGN DAYS).

Rot.	Estimator	Bal-acc	Spec.	Recall
R-Z	Random forest	0.8603	0.7231	0.9974
	CatBoost	0.8218	0.6462	0.9974
	XGBoost	0.8385	0.6769	1.0000
R-M	Random forest	0.8529	0.9077	0.7980
	CatBoost	0.8617	0.9154	0.8079
	XGBoost	0.8353	0.8923	0.7782
R-T	Random forest	0.7973	0.7308	0.8638
	CatBoost	0.7520	0.6615	0.8426
	XGBoost	0.7478	0.6615	0.8340
R-S	Random forest	0.8346	0.7154	0.9538
	CatBoost	0.8487	0.7154	0.9821
	XGBoost	0.8038	0.6615	0.9462

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TABLE XIX
EXPLANATION QUALITY FOR EVERY SCORED PREDICTION SET ($n = 1{,}520$ MANIPULATED DAYS; PARAMETER EXTRACTION ON THE FIVE ATTACKS WITH A STATED PARAMETER, $n = 610$).

Run	ROUGE-L	Param. mentioned	Param. correct
R0 (label only)	0.000	0.000	0.000
R1 (rationale+label)	0.454	0.953	0.798
Qwen3-8B (workhorse)	0.455	0.949	0.775
Qwen3-14B	0.456	0.948	0.682
Phi-4-reasoning	0.457	**0.969**	**0.802**
Ministral-3-14B	**0.462**	0.951	0.669
Llama-3.1-8B	0.453	0.949	0.671
Gemma-4-E4B	0.444	0.892	0.712
Gemma-4-31B	0.431	0.890	0.666
Qwen3-8B (magnitude)	0.458	0.966	0.795

TABLE XX
TRAINING COST PER RUN: TRAINING ROWS, WALL-CLOCK, AND FINAL TRAINING LOSS.

Run	Train rows	Wall-clock (h)	Final loss
Qwen3-8B (workhorse)	6,589	5.23	0.407
Qwen3-14B	6,589	8.29	0.387
Phi-4-reasoning	6,589	5.52	0.424
Ministral-3-14B	6,589	7.52	0.354
Llama-3.1-8B	6,589	4.12	0.410
Gemma-4-E4B	6,589	7.43	0.338
Gemma-4-31B (4-bit)	6,589	22.36	0.373
Qwen3-8B (magnitude)	6,589	5.43	0.371
$r = 8$ / $r = 32$	6,589	5.24 / 5.23	0.406 / 0.408
R-Z / R-M / R-T / R-S	5,035–5,646	3.6–4.5	0.39–0.44
R0 / R1	6,589	4.62 / 4.73	0.014 / 0.424
$N \in \{65, 260, 1{,}040\}$	65–1,040	0.05–0.83	0.56–1.25

R0's low loss (0.014) reflects label-only targets being near-trivially learnable; the gap to R1 reflects the cost of the rationale tokens rather than model quality.

TABLE XXI
MAGNITUDE MAE PER ATTACK (MAGNITUDE-BEARING ATTACKS; n PER CLASS AS IN SECTION III-A).

Attack	MAE (ρ)
attack0	0.0000
attack12	0.0302
attack3	0.0320
attack5	0.0335
attack7	0.0374
attack1	0.0447
attack13	0.0468
attack6	0.0607
attack4	0.0622
attack2	0.0653

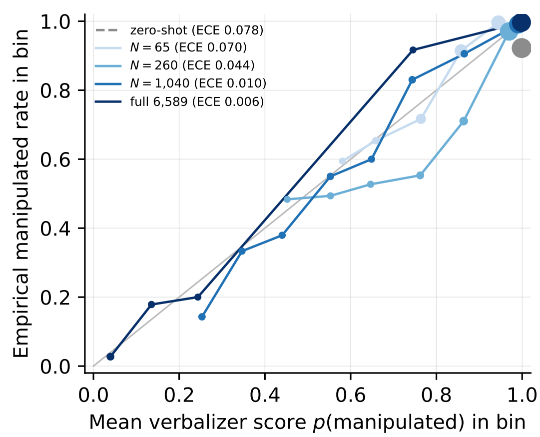

Fig. 8. Reliability of the verbalizer score across training stages (ten equal-width bins on the evaluation slice; multi-seed stages show the seed-mean curve). Bins holding at least ten days are drawn, with marker area tracking bin support; the expected calibration error in the legend is computed over all bins, support-weighted. Discrimination precedes calibration: the AUC is high from the start while the calibration gap closes only with the full training data.

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